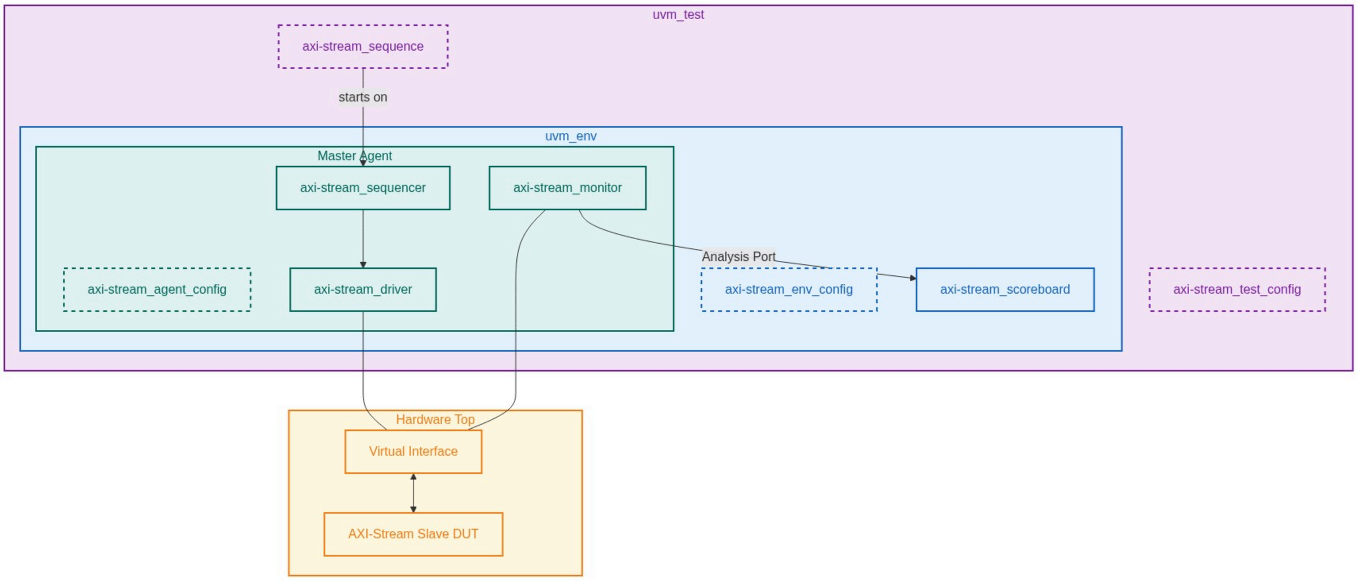
Verification Plan Report

Project: AXI-Stream Slave DUT

Version: 3.0

1. UVM Environment Architecture



2. Test Requirements

Index	Name	Description
REQ_HANDSHAKE_01	TVALID/TREADY Handshake Rules	Verification of the 2-way flow control. TVALID must not wait for TREADY. TVALID must remain stable until TREADY is asserted.
REQ_BYTE_QUAL_01	TKEEP and TSTRB Handling	Correct identification of valid bytes (TKEEP) and data vs position bytes (TSTRB).
REQ_PARITY_01	Odd Parity Protection	Identify single-bit flips on TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, and TUSER.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_RAND_PKT_SIZE	Send packets with randomized beat counts and TREADY backpressure.	REQ_HANDSHAKE_01
TC_SPARSE_STRM	Verify slave can handle sparse streams with intermittent TKEEP/TSTRB bits.	REQ_BYTE_QUAL_01
TC_PARITY_ERR	Inject odd parity errors on TDATACHK and verify error signal detection.	REQ_PARITY_01

4. Functional Coverage Groups

Group Name	Description	TR Mapping
cg_handshake	Coverage for TVALID/TREADY being HIGH/LOW simultaneously.	REQ_HANDSHAKE_01
cg_data_width	Ensure all data bits in TDATA have toggled.	REQ_BYTE_QUAL_01

7. Interface Signals

Signal Name	Direction	Width	Description
ACLK	In	1	Clock
TVALID	In	1	Transmit valid
TREADY	Out	1	Receive ready
TDATA	In	DATA_WIDTH	Payload
TDATACHK	In	DATA_WIDTH/8	Parity bits (AXI5)

8. Verification Environment Structure

Root: axi_strm_slv_env/

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|-- agent : Master driving Slave
|-- env : UVM Environment
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